



# Qasim Hussain

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## SKILLS

### Wet Lab

Cell Culture (cell freezing, cell thawing, cell passaging, cell counting and cloning) | Virus Culture | *In Vitro* Assays (cell viability assay, LDH assay, virus inhibition assay, G/F actin assay) | RNA and DNA Extraction | RT-qPCR | Gel Electrophoresis | Bacterial Culture | Protein production and purification | SDS PAGE | Western Blot | Confocal Microscopy | Serum biochemistry (AST, ALT, triglycerides, total cholesterol and glucose) | Handling, restraint, and physiological maintenance of murine (mice) and avian (pigeon) experimental models | Compliance with animal welfare protocols and execution of controlled dosing regimens, etc.

### Dry Lab (NGS Analysis)

Whole Genome Sequencing (Short Reads, Long Reads, Hybrid) | Whole Exome Sequencing | Bulk RNA-Seq | Single Cell RNA-Seq | Spatial Transcriptomics | Pangenomics | Antibiotic Resistance Genes Detection (AMR) Analysis | CAZyme Analysis | Secretion System Detection | CRISPR-Cas | COJScan and TFFscan System Analysis | Biosynthetic Gene Cluster Analysis | ImmunoPhagomics | WGCNA | PanGWAS | TWAS | Ribo-Seq | Amplicon Sequencing (16S, 18S and ITS) | Metagenomics, etc.

### Dry Lab (Structural Bioinformatics)

Machine Learning | Molecular Docking | Molecular Dynamics Simulation | Virtual Screening | Protein Structure Prediction and Modeling | *De Novo* Protein Design | *De Novo* Binder Prediction | *De Novo* Aptamers Prediction, etc.

### Programming & Workflow Management

Markdown | Python | R | Linux/Bash | SQL | Git | Conda | Snakemake, etc.

## EDUCATION & TRAINING

[ 9 Sep 2024 - 10 Jul 2026 ]

### MS Animal Vaccine Technology

*National Pingtung University of Science and Technology*

**Final grade:** 93.63% | **Thesis:** *In silico* drug discovery via high-throughput virtual screening and *in vitro* validation of novel antivirals of Porcine Epidemic Diarrhea Virus (PEDV).

[ 4 Sep 2019 - 31 Aug 2023 ]

### BS Zoology

*University of Education, Township, Lahore*

## WORK EXPERIENCE

[ 13 Aug 2026 - Current ]

*Mackay Memorial Hospital*

### Research Assistant (Full-time) | Cardiovascular Research Group

[ 9 Sep 2024 - 10 Jul 2026 ]

*National Pingtung University of Science and Technology*

### Graduate Researcher | Graduate Institute of Animal Vaccine Technology

[ 21 Dec 2025 - Current ]

*Portfolio*

### Independent Computational Research

## PUBLICATIONS

1. [Lee, C.M., Villanueva, B.H.A., Minh, H., Hussain, Q.\\*, & Chuang, K.-P.\\* \(2025\). Evaluation of feline exosome mediated renal regeneration in adenine-induced chronic kidney disease. \[co-corresponding author\]](#)
2. [Villanueva, B.H.A., Gomez, M.C., Hussain, Q., Hong, Q.D., & Chuang, K.-P. \(2025\). Conserved internal PB2-PB1-PA-NP-M genes and divergent HA/NA in bat influenza A viruses H9N2, H17N10, and H18N11.](#)

3. Villanueva, B.H.A., [Hussain, Q.](#), Hong, Q.D., Chen, M.H., Minh, H., Huang, H.Y., Tyan, Y.C., & Chuang, K.P. Emergence of Black Queen Cell Virus in *Apis mellifera ligustica* of southern Taiwan: evidence from local genomic surveillance. ([Under Review](#))
4. [Nadeem, S.](#), [Tahir, S.B.](#), [Ahmad, M.](#), [Aslam, A.](#), [Farooq, M.](#), [Hussain, Q.](#), [Majid, H.](#), [Usman, S.M.](#), [Bashir, S.](#), & [Magsood, H.](#) (2023). The next pandemic: a comprehensive look into zoonotic emerging threats. ([Book Chapter](#))
5. Afzal, H., Umar, M., [Hussain, Q.](#), & Cheng, L.-T.\* Evaluating the adjuvanticity of N-D1 domain of Flagellin encompassing hot-spot region in combination with Porcine Epidemic Diarrhoea Virus (PEDV) ([Under Review](#))
6. Are *de novo* designed proteins immunogenic? ([Review: Under Preparation](#))

## CONFERENCES & SEMINARS

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1. Umar, M., [Hussain, Q.](#), & Cheng, L.-T.\* "From coexistence to contagion: assessing zoonotic threats from Formosan Macaques in urban Taiwan." 2025 Grand Challenge Taiwan SDGs Issue English Poster Competition, NCKU. ([Poster, 2nd author](#))
  2. [Hussain, Q.](#), & Chuang, K.P.\* "*In vitro* antiviral efficacy of Molnupiravir against Porcine Epidemic Diarrhea Virus (PEDV)." 5th International Symposium on Immunotechnology and Animal Biologics Development, NPUST. ([Poster, 1st author](#))
  3. Duong, H.Q., [Hussain, Q.](#), Nguyen, D.L., & Chuang, K.-P.\* "Potential application of plant-derived nanoparticles as antiviral agents against Parvovirus." 2026 International Nanotechnology Olympiad, Academia Sinica. ([Poster, 2nd author](#))
  4. [Hussain, Q.](#), Villanueva, B.H.A., Chuang, K.P.\*, & Hsin, K.T.\* "*In silico* drug discovery via high-throughput virtual screening and *in vitro* validation of novel antivirals against Porcine Epidemic Diarrhea Virus (PEDV) Spike Protein." 2026 International Symposium on Sustainable Innovation in Smart Agriculture (ISSI-SA 2026), NPUST. ([Poster, 1st author](#))
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## HONOURS AND AWARDS

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|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| [ 13 Jun 2026 ] | <b>Outstanding Researcher Award</b> ( <b>Awarding institution:</b> National Pingtung University of Science and Technology)              |
| [ 17 Jan 2026 ] | <b>Chung Hwa Rotary Education Foundation Scholarship Holder</b> ( <b>Awarding institution:</b> Chung Hwa Rotary Educational Foundation) |
| [ 4 Sep 2019 ]  | <b>University Scholarship Holder</b> ( <b>Awarding institution:</b> University of Education, Lahore, Pakistan)                          |
| [ 21 Jul 2017 ] | <b>College scholarship holder</b> ( <b>Awarding institution:</b> Webster Group of Colleges)                                             |
| [ 11 Sep 2017 ] | <b>Prime Minister Laptop</b> ( <b>Awarding institution:</b> Punjab Board of Technical Education)                                        |
| [ 11 May 2015 ] | <b>Merit Scholarship Holder</b> ( <b>Awarding institution:</b> Punjab Educational Endowment Fund (PEEF))                                |

## LANGUAGE SKILLS

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English ([IELTS: In Preperation](#))

## RECOMMENDATIONS

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Will be provided on demand.